

EE 473 - HOMEWORK 2

Boran Aybak Kiliç
2022401207

$$X(\omega) = \frac{1}{1 - a e^{-j\omega}} \xleftrightarrow{\mathcal{F}^{-1}} x(n) = a^n u(n) \quad (\text{assuming it's causal})$$

a) 1st Method

Using the time domain signal, calculate $y(n) = x(2n+1) = a^{2n+1} u(2n+1)$

$$Y(\omega) = \sum_{n=-\infty}^{\infty} a^{2n+1} u(2n+1) e^{-j\omega n} = a \sum_{n=0}^{\infty} (a^2 e^{-j\omega})^n = \boxed{\frac{a}{1 - a^2 e^{-j\omega}}} \quad \text{solution manual is wrong}$$

→ Although this is straightforward, there is a more elegant solution

2nd Method

Let $v(n) = x(n+1)$, Then $V(\omega) = e^{+j\omega} X(\omega)$ since $\mathcal{F}\{x(n-n_0)\} = e^{-j\omega n_0} X(\omega)$

$y(n) = v(2n) \Rightarrow$ We want to find how their Fourier transforms relate

Define $w(n) = \begin{cases} v(n), & \text{if } n \text{ is even} \\ 0, & \text{if } n \text{ is odd} \end{cases} \Rightarrow v$

$$Y(\omega) = \sum_{n=-\infty}^{\infty} y(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} v(2n) e^{-j\omega n}$$

$$W(\Omega) = \sum_{m=-\infty}^{\infty} w(m) e^{-j\Omega m} \xrightarrow{\Omega = \frac{\omega}{2}} W\left(\frac{\omega}{2}\right) = \sum_{m=-\infty}^{\infty} w(m) e^{-j\frac{\omega}{2} m}$$

Since $w(m)$ is zero for odd values by definition, we only need to sum over even values, $m = 2n$

$$W\left(\frac{\omega}{2}\right) = \sum_{n=-\infty}^{\infty} w(2n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} v(2n) e^{-j\omega n} = Y(\omega)$$

↳ $v(2n) = w(2n)$ by definition

→ Now we need to relate $V(\omega)$ to $W(\omega)$.

Notice that $w(n) = v(n) \left[\frac{1+(-1)^n}{2} \right] = \frac{1}{2} v(n) + \frac{1}{2} v(n) e^{-j\pi n}$

$$W(\omega) = \frac{1}{2} V(\omega) + \frac{1}{2} V(\omega - \pi)$$

$$Y(\omega) = \frac{1}{2} \left[V\left(\frac{\omega}{2}\right) + V\left(\frac{\omega}{2} - \pi\right) \right] = \frac{1}{2} \left[\frac{e^{+j\omega/2}}{1 - a e^{-j\omega/2}} + \frac{e^{+j(\frac{\omega}{2} - \pi)}}{1 - a e^{-j(\frac{\omega}{2} - \pi)}} \right]$$

$$= \frac{e^{+j\omega/2}}{2} \left[\frac{1}{1 - a e^{-j\omega/2}} - \frac{1}{1 + a e^{-j\omega/2}} \right] = \boxed{\frac{a}{1 - a^2 e^{-j\omega}}} \quad \text{in accordance with 1st method.}$$

b) $y(n) = e^{\pi n/2} x(n+2) = e^{j(j\pi/2)n} x(n+2)$

$$Y(\omega) = \mathcal{F}\{X(\omega) e^{j\omega 2}\}(\omega + j\pi/2) \quad \text{using time and frequency shifting properties.}$$

$$Y(\omega) = \left\{ \frac{e^{j\omega 2}}{1 - a e^{-j\omega}} \right\}(\omega + j\pi/2) = \frac{e^{j2(\omega + j\pi/2)}}{1 - a e^{-j(\omega + j\pi/2)}} = \frac{-e^{j2\omega} e^{-\pi}}{1 - a e^{+j\pi/2} e^{-j\omega}}$$

! Note that $e^{\pi n/2}$ does not make much sense since $\lim_{n \rightarrow \infty} n(n) = \infty$, and thus, the signal actually "does not have a Fourier transform. Therefore, it's not too absurd to assume there was a typo in the question and $y(n) = e^{-j\pi n/2} x(n+2)$ in that case!

$$Y(\omega) = \mathcal{F}\{X(\omega) e^{j\omega 2}\}(\omega - \pi/2) = \frac{e^{j2(\omega - \pi/2)}}{1 - a e^{-j(\omega - \pi/2)}} = \boxed{\frac{-e^{j2\omega}}{1 - a e^{-j\omega}}} \quad \text{corrected signal}$$

c) Using the result in a) and $x(n) \leftrightarrow X(\omega)$ [time-reversal property]

$$x(2n) \xrightarrow{\mathcal{F}} \frac{1}{2} \left[\frac{1}{1 - a e^{-j\omega/2}} + \frac{1}{1 + a e^{-j\omega/2}} \right] = \frac{1}{1 - a^2 e^{-j\omega}}$$

$$Y(\omega) = \mathcal{F}\{x(-2n)\} = \boxed{\frac{1}{1 - a^2 e^{-j\omega}}}$$

d)

$$y(n) = x(n) \cos(0.3\pi n) = \frac{1}{2} x(n) e^{j0.3\pi n} + \frac{1}{2} x(n) e^{-j0.3\pi n}$$

$$Y(\omega) = \frac{1}{2} X(\omega - 0.3\pi) + \frac{1}{2} X(\omega + 0.3\pi) \quad \text{* using freq. shift property}$$

$$= \frac{1}{2} \left[\frac{1}{1 - ae^{-j\omega} e^{j0.3\pi}} + \frac{1}{1 - ae^{-j\omega} e^{-j0.3\pi}} \right] = \frac{1}{2} \frac{2 - ae^{-j\omega} 2 \cos(0.3\pi)}{1 - 2ae^{-j\omega} \cos(0.3\pi) + a^2 e^{-j2\omega}}$$

$$= \boxed{\frac{1 - ae^{-j\omega} \cos(0.3\pi)}{1 - 2ae^{-j\omega} \cos(0.3\pi) + a^2 e^{-j2\omega}}}$$

$$e) \quad y(n) = x(n) * x(n-1) \xrightarrow{\mathcal{F}} [X(\omega)] \cdot [X(\omega) e^{-j\omega}] = Y(\omega)$$

$$Y(\omega) = X^2(\omega) e^{-j\omega} = \boxed{\frac{e^{-j\omega}}{(1 - ae^{-j\omega})^2}}$$

$$f) \quad y(n) = x(n) * x(-n) \xrightarrow{\mathcal{F}} X(\omega) \cdot X(-\omega) = Y(\omega)$$

$$Y(\omega) = \frac{1}{1 - ae^{-j\omega}} \cdot \frac{1}{1 - ae^{j\omega}} = \boxed{\frac{1}{1 - 2a \cos \omega + a^2}}$$

$$2) \quad x(n) = \{1, 0, -1, \underset{\uparrow}{2}, 3\} \quad \text{with} \quad X(\omega) = X_R(\omega) + j X_I(\omega)$$

Let us decompose the signal into its even and odd parts.

$$X_e(n) = \frac{x(n) + x(-n)}{2} = \left\{ \frac{1}{2}, 0, 1, \underset{\uparrow}{2}, 1, 0, \frac{1}{2} \right\}$$

$$X_o(n) = \frac{x(n) - x(-n)}{2} = \left\{ \frac{1}{2}, 0, -2, \underset{\uparrow}{0}, 2, 0, -\frac{1}{2} \right\}$$

Note that even part of a real signal corresponds to the even and real part of the Fourier transform and odd part corresponds to even and imaginary part of its FT.

$$\text{Then } X_R(\omega) = \sum_{n=-3}^3 x_e(n) e^{-j\omega n}, \quad jX_I(\omega) = \sum_{n=-3}^3 x_o(n) e^{-j\omega n}$$

For $Y(\omega) = X_I(\omega) + X_R(\omega) e^{j2\omega}$, we have

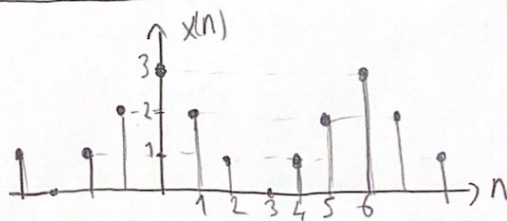
$$y(n) = \mathcal{F}^{-1}\{Y(\omega)\} = \underbrace{\mathcal{F}^{-1}\{X_I(\omega)\}}_{\frac{x_o(n)}{j}} + \underbrace{\mathcal{F}^{-1}\{X_R(\omega)\}}_{x_e(n)} (n+2)$$

$$= -j x_o(n) + x_e(n+2) = \left\{ \frac{1}{2}, 0, 1-j\frac{1}{2}, 2, 1+2j, 0, \frac{1}{2}-2j, 0, j\frac{1}{2} \right\}$$

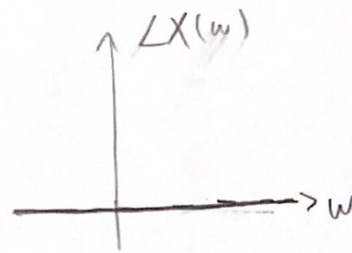
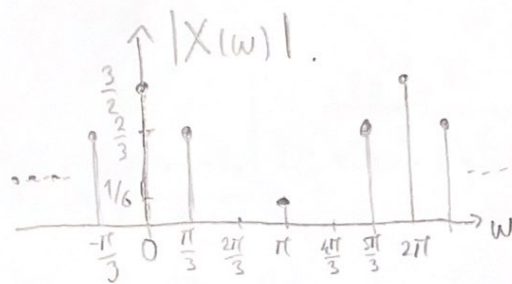
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 $-5 \quad -4 \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3$

3)

$$x(n) = \{-1, 0, 1, 2, 3, 2, 1, 0, 1, 2, 3, 2, \dots\}$$



$$c_0 = \frac{3}{2}, c_1 = \frac{2}{3}, c_2 = 0, c_3 = \frac{1}{6}, c_4 = 0, c_5 = \frac{2}{3}$$



b) Parseval's Theorem: $\frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = \sum_{k=0}^{N-1} |c_k|^2$

$$\frac{1}{6} (3^2 + 2^2 + 1^2 + 1^2 + 2^2) = \frac{19}{6} = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \xleftarrow{\text{EQUAL}} \sum_{k=0}^{N-1} |c_k|^2$$

$$\left(\frac{3}{2}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{1}{6}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{9}{4} + \frac{4}{9} + \frac{1}{36} + \frac{4}{9} = \frac{81 + 16 + 1 + 16}{36} = \frac{114}{36} = \frac{19}{6} = \sum_{k=0}^{N-1} |c_k|^2$$

4) $X(z) = \frac{1 + 2z^{-1}}{1 - 2z^{-1} + z^{-2}}$

a) If $x(n)$ is causal ROC is exterior of a circle. Thus we seek a solution in negative powers of z with descending order,

$$\begin{array}{r} 1 + 2z^{-1} \bigg| 1 - 2z^{-1} + z^{-2} \\ - 1 - 2z^{-1} + z^{-2} \quad 1 + 4z^{-1} + 7z^{-2} + 10z^{-3} + 13z^{-4} \dots \\ \hline 4z^{-1} - z^{-2} \\ - 4z^{-1} - 8z^{-2} + 4z^{-3} \\ \hline 0 + 7z^{-2} - 4z^{-3} \\ 7z^{-2} - 14z^{-3} + 7z^{-4} \\ \hline 10z^{-3} - 7z^{-4} \\ 10z^{-3} - 20z^{-4} + 10z^{-5} \\ \hline 13z^{-4} - 10z^{-5} \end{array}$$

$$\begin{aligned} X(z) &= 1 + 4z^{-1} + 7z^{-2} + 10z^{-3} + 13z^{-4} + \dots \\ &= \sum_{k=0}^{\infty} (1+3k)z^{-k} \end{aligned}$$

$$x(n) = \{0, \underset{\uparrow}{1}, 4, 7, 10, 13, 16, 19, 22, \dots\}$$

$$x(n) = (3n+1)u(n)$$

b) If $x(n)$ is anticausal, ROC is interior of a circle. $X(z)$ will comprise of positive, ascending powers of z .

$$\begin{array}{r} 2z^{-1} + 1 \overline{) z^{-2} - 2z^{-1} + 1} \\ \underline{2z^{-1} - 4 + 2z} \\ 5 - 2z^{-1} \\ \underline{5 - 10z + 5z^2} \\ 8z - 5z^2 \\ \underline{8z - 16z^2 + 8z^3} \\ 11z^2 - 8z^3 \end{array}$$

$$\Rightarrow X(z) = 2z + 5z^2 + 8z^3 + 11z^4 + \dots$$

$$x(n) = 2\delta(n+1) + 5\delta(n+2) + 8\delta(n+3) + \dots$$

$$\boxed{x(n) = \{\dots, 11, 8, 5, 2, 0, 0, 0, \dots\}} \\ \uparrow \\ = (-3n-1)u(-n-1)$$

$$5) X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

By partial fraction expansion: $X(z) = \frac{A}{1-2z^{-1}} + \frac{B}{3-z^{-1}}$

$$\left. \begin{array}{l} 3A+B=0 \\ -A-2B=5 \end{array} \right\} \begin{array}{l} A=1 \\ B=-3 \end{array} \left\} X(z) = \frac{1}{1-2z^{-1}} - \frac{3}{3-z^{-1}}$$

We know that the causal signal $x_c(n) = a^n u(n)$ has the Z-transform

$$\mathcal{Z}\{a^n u(n)\} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1-az^{-1}} \quad \text{with } |az^{-1}| < 1 \Rightarrow \frac{\text{ROC}}{|z| > |a|}$$

On the other hand anti-causal signal $x_a(n) = -a^n u(-n-1)$ has Z-transform

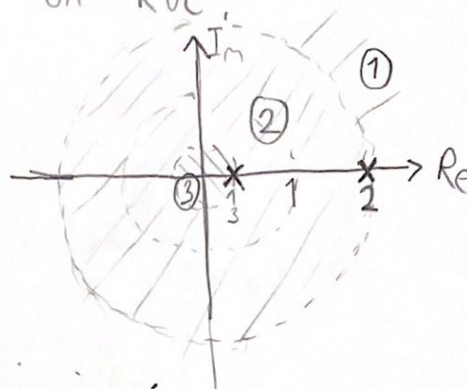
$$\begin{aligned} \mathcal{Z}\{-a^n u(-n-1)\} &= -\sum_{n=-\infty}^{-1} a^n z^{-n} = -\sum_{n=1}^{\infty} a^{-n} z^n = -\left[\sum_{n=0}^{\infty} (a^{-1}z)^n - 1 \right] \\ &= -\left[\frac{1}{1-a^{-1}z} - 1 \right] = \frac{-a^{-1}z}{1-a^{-1}z} = \frac{-1}{az^{-1}-1} = \frac{1}{1-az^{-1}} \end{aligned}$$

$$\text{with } |a^{-1}z| < 1 \Rightarrow \frac{\text{ROC}}{|z| < |a|}$$

We conclude that 2 signals having the same Z-transform can have different inverse transform based on ROC.

$$X(z) = \frac{1}{1-2z^{-1}} - \frac{1}{1-\frac{1}{3}z^{-1}}$$

* There are 3 possible ROC's
X(z) can belong to



1) $|z| > 2$: Causal $x(n)$

$$x(n) = 2^n u(n) - \left(\frac{1}{3}\right)^n u(n)$$

2) $\frac{1}{3} < |z| < 2$: Stable $x(n) \Rightarrow$ Case where ROC is outward of $z = \frac{1}{3}$ but inwards of $z = 2$
We will have causal $\left(\frac{1}{3}\right)^n u(n)$ and anticausal $-2^n u(-n-1)$

$$x(n) = -2^n u(-n-1) - \left(\frac{1}{3}\right)^n u(n)$$

3) $|z| < \frac{1}{3}$: Anticausal $x(n)$

$$x(n) = -2^n u(-n-1) + \left(\frac{1}{3}\right)^n u(-n-1)$$

6) $y(n) = -a_1 y(n-1) + b_0 x(n) + b_1 x(n-1)$

a) 1st Method

Let $x(n) = \delta(n)$ and let $y(n) = 0$ for $n < 0$ since there is no input before $n=0$, it's plausible to expect zero-state.

$$\left. \begin{aligned} y(0) &= -a_1 y(-1) + b_0 \delta(0) + b_1 \delta(-1) = b_0 \\ y(1) &= -a_1 y(0) + b_1 = b_1 - a_1 b_0 \\ y(2) &= -a_1 y(1) = (b_1 - a_1 b_0)(-a_1) \\ y(3) &= -a_1 y(2) = (b_1 - a_1 b_0)(-a_1)^2 \\ &\vdots \\ y(k) &= -a_1 y(k-1) = (b_1 - a_1 b_0)(-a_1)^{k-1} \end{aligned} \right\} \begin{aligned} h(n) = y(n) &= b_0 \delta(n) \\ &+ (b_1 - a_1 b_0)(-a_1)^{n-1} u(n-1) \end{aligned}$$

6.a) 2nd Method

Taking the Z-transform of difference eq.

$$Y(z) = -a_1 z^{-1} Y(z) + b_0 X(z) + b_1 z^{-1} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}} = \frac{b_0}{1 + a_1 z^{-1}} + b_1 z^{-1} \frac{1}{1 + a_1 z^{-1}}$$

$$\begin{aligned} h(n) &= \mathcal{Z}^{-1}\{H(z)\} = b_0 a_1^n u(n) + b_1 a_1^{n-1} u(n-1) \\ &= b_0 \delta(n) + (b_1 - b_0 a_1) (-a_1)^n u(n-1) \end{aligned}$$

b) For $x(n) = u(n) \Rightarrow s(n) = h(n) * x(n) = h(n) * u(n)$: step response

$$S(z) = H(z) \mathcal{Z}\{u(n)\} = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}} \cdot \frac{1}{1 - z^{-1}}$$

By partial fraction expansion : $S(z) = \frac{A}{1 + a_1 z^{-1}} + \frac{B}{1 - z^{-1}}$

$$\left. \begin{aligned} A + B &= b_0 \\ -A + a_1 B &= b_1 \end{aligned} \right\} \begin{aligned} B &= \frac{b_0 + b_1}{1 + a_1} \\ A &= \frac{b_0 a_1 - b_1}{1 + a_1} \end{aligned} \left\{ S(z) = \frac{b_0 a_1 - b_1}{1 + a_1} \frac{1}{1 + a_1 z^{-1}} + \frac{b_0 + b_1}{1 + a_1} \frac{1}{1 - z^{-1}} \right.$$

$$s(n) = \left[\frac{b_0 + b_1}{1 + a_1} + \frac{b_0 a_1 - b_1}{1 + a_1} (-a_1)^n \right] u(n)$$

c) For $y(-1) = A \neq 0$, we can't use the bilateral z-transform anymore. We need to opt for unilateral z-transform

$$\begin{aligned} Y^+(z) &= \sum_{n=0}^{\infty} y(n) z^{-n}, \quad \mathcal{Z}^+\{y(n-1)\} = z^{-k} \left\{ \sum_{l=-1}^{-1} x(l) z^{-l} + \sum_{l=0}^{\infty} x(l) z^{-l} \right\} \\ &= z^{-1} Y(z) + y(-1) \end{aligned}$$

Then taking unilateral z-transform of both sides:

$$Y^+(z) + a_1 [z^{-1} Y^+(z) + y(-1)] = b_0 X^+(z) + b_1 [z^{-1} X^+(z) + x(-1)]$$

Substituting $y(-1) = A$, $x(n) = u(n)$ & $x(-1) = 0$

$$Y^+(z) = \underbrace{\frac{b_0 + z^{-1} b_1}{1 + a_1 z^{-1}} X^+(z)}_{\text{Zero-state response}} + \underbrace{\frac{a_1 A}{1 + a_1 z^{-1}}}_{\text{zero-input response}}$$

$$y(n) = \underbrace{\left[\frac{a_1 b_0 - b_1}{1 + a_1} (-a_1)^n + \frac{b_0 + b_1}{1 + a_1} \right] u(n)}_{y_{zs}(n)} - \underbrace{a_1 A (-a_1)^n u(n)}_{y_{zi}(n)}$$

$$\begin{aligned} d) \quad x(n) &= \cos(\omega_0 n) u(n), \quad X(z) = \frac{1}{2} \sum_{n=0}^{\infty} (e^{j\omega_0 n} + e^{-j\omega_0 n}) z^{-n} = \frac{1/2}{1 - e^{j\omega_0} z^{-1}} + \frac{1/2}{1 - e^{-j\omega_0} z^{-1}} \\ &= \frac{1}{2} \frac{2 - 2z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}} = \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}} \end{aligned}$$

$$Y(z) = X(z) H(z) = \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}} \cdot \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}} = \frac{b_0 + (b_1 - b_0 \cos \omega_0) z^{-1} - b_1 \cos \omega_0 z^{-2}}{[1 + a_1 z^{-1}][1 - 2z^{-1} \cos \omega_0 + z^{-2}]}$$

By partial fraction exp: $Y(z) = \frac{A}{1 + a_1 z^{-1}} + \frac{B}{1 - e^{j\omega_0} z^{-1}} + \frac{C}{1 - e^{-j\omega_0} z^{-1}}$

$$A = \left[(1 + a_1 z^{-1}) Y(z) \right]_{z = -\frac{1}{a_1}} = \frac{[b_0 + b_1 (-\frac{1}{a_1})] [1 - (-\frac{1}{a_1}) \cos \omega_0]}{(1 + e^{\frac{j\omega_0}{a_1}}) (1 + \frac{e^{-j\omega_0}}{a_1})}$$

$$= \frac{(a_1 b_0 - b_1)(a_1 + \cos \omega_0)}{(a_1 + e^{j\omega_0})(a_1 + e^{-j\omega_0})} = \frac{(a_1 b_0 - b_1)(a_1 + \cos \omega_0)}{a_1^2 + 2a_1 \cos \omega_0 + 1}$$

$$B = \left[(1 - e^{j\omega_0} z^{-1}) Y(z) \right]_{z^{-1} = e^{-j\omega_0}} = \frac{(b_0 + b_1 e^{-j\omega_0})(1 - \cos \omega_0 e^{-j\omega_0})}{(1 + a_1 e^{-j\omega_0})(1 - e^{-2j\omega_0})}$$

$$= \frac{(b_0 + b_1 e^{-j\omega_0}) \left[1 - \left(\frac{1}{2} e^{j\omega_0} e^{-j\omega_0} + \frac{1}{2} e^{j\omega_0} e^{-j\omega_0} \right) \right]}{[]} = \frac{1}{2} \left(\frac{b_0 + b_1 e^{-j\omega_0}}{1 + a_1 e^{-j\omega_0}} \right)$$

$$C = H(e^{-j\omega_0}) = \frac{1}{2} H^*(e^{j\omega_0}) = B^* = \frac{1}{2} H(e^{j\omega_0})$$

$$Y(z) = \frac{A}{1 + a_1 z^{-1}} + \frac{B}{1 - e^{j\omega_0} z^{-1}} + \frac{C}{1 - e^{-j\omega_0} z^{-1}}$$

$$y(n) = \underbrace{A (-a_1)^n u(n)}_{y_{\text{transient}}(n)} + \underbrace{B (e^{j\omega_0})^n u(n) + [B (e^{j\omega_0})^n]^* u(n)}_{y_{\text{ss}}(n)}$$

$$y_{\text{ss}}(n) = 2 \operatorname{Re} \{ B e^{j\omega_0 n} \} = \operatorname{Re} \{ H(e^{j\omega_0}) e^{j\omega_0 n} \}$$

$$= |H(\omega_0)| \cos(\omega_0 n + \angle H(\omega_0))$$

$$y(n) = \left[A (-a_1)^n + |H(\omega_0)| \cos(\omega_0 n + \angle H(\omega_0)) \right] u(n)$$

where $A = \frac{(a_1 b_0 - b_1)(a_1 + \cos \omega_0)}{1 + a_1^2 + 2a_1 \cos \omega_0}$

$$|H(\omega_0)| = \frac{\sqrt{b_0^2 + b_1^2 + 2b_1 b_0 \cos \omega_0}}{\sqrt{1 + a_1^2 + 2a_1 \cos \omega_0}}$$

$$\angle H(\omega_0) = \tan^{-1} \left(\frac{-b_1 \sin \omega_0}{b_0 + b_1 \cos \omega_0} \right) - \tan^{-1} \left(\frac{-a_1 \sin \omega_0}{1 + a_1 \cos \omega_0} \right)$$